



Society of Petroleum Engineers

SPE-229269-MS

Achieving Road Safety Through Implementation of Enhanced Vehicle Safety Requirements (Re-Defining Vehicle Critical Safety Elements)

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

Driving-related incidents have historically been the leading cause of fatalities in the oil and gas industry, accounting for approximately one in every three deaths (IOGP Report 365). With driver error contributing to over 70% of major road transport incidents, managing driver behavior and road transport risk has become a top priority across the industry.

Road safety management remains complex due to the interplay of numerous risk factors, driver competence and condition, vehicle suitability, road conditions, and environmental influences. While many controls traditionally used to manage these risks are administrative in nature and thus less effective according to the hierarchy of controls, technological advancements have introduced more robust engineering and system-based solutions. Key vehicle engineering controls like Anti-lock Braking Systems (ABS), Electronic Stability Control (ESC), Rollover Prevention Systems (RPS), and Active Fatigue Detection Device (AFDD) linked with IVMS now work in synergy to reduce both the likelihood and severity of incidents. These systems offer both proactive and reactive interventions, mitigating human error and enhancing recovery when incidents do occur.

This paper outlines how Petroleum Development Oman (PDO) has embraced vehicle technology advancements as part of its integrated road safety strategy. Through the redefinition of vehicle safety specifications and a structured implementation plan, PDO has driven significant improvement in its transport safety performance. Key to this transformation has been strong collaboration with internal and external stakeholders and the enforcement of updated vehicle standards. It is worth highlighting that PDO's road safety Improvement program, called "Darb Al Salama," considers safe vehicles as one of the key safety pillars, alongside safe drivers, safe roads, and enablers such as digitalization and safety management. Launched in 2021, the Darb Al Salama program has successfully led to improvements in vehicle safety during this period.

Since the introduction of Darb Al Salama and update of the road safety standard in early 2022, vehicle safety rejuvenation with new contemporary features has risen from 2% to 36%, with a target of reaching 100% by 2030. Furthermore, severe motor vehicle incidents (MVIs) linked to vehicle-related causes have decreased by 62% between 2021 and 2025. This paper shares insights into PDO's journey, the challenges

faced, and the tangible impact of integrating advanced vehicle technologies on road transport safety in the oil and gas sector.

Introduction

Road transport safety continues to be a paramount concern in the oil and gas industry, where driving-related incidents have persistently accounted for the leading cause of fatalities worldwide (International Association of Oil & Gas Producers [IOGP], 2020). Petroleum Development Oman (PDO) operates across an extensive concession area of approximately 90,000 square kilometers in the Sultanate of Oman. This vast operational footprint includes challenging environments characterized by extreme desert conditions, where summer temperatures often exceed 50°C in some areas, and a significant proportion of roadways are unpaved or graded without formal infrastructure. Managing road safety in such a context is further complicated by the sheer scale of operations, which include over 300 active contractor agreements, more than 10,000 vehicles, and an extensive driver population exceeding 20,000 individuals covering average of 320 million KM annually. These factors contribute to a complex safety management landscape that demands a robust, integrated approach.

Effective road safety management within PDO has been strategically organized around three fundamental pillars, which collectively address the multifaceted nature of road transport risks, adopting UN Road Safety Decade of action 2021-2030, concept on "Safe System Approach" where Safe vehicles is one of the key pillars

- a. **Driver Pillar:** Focused on driver behavior, competence, and condition, addressing human factors that are known to contribute to over 70% of major road transport incidents both globally and within PDO operations (IOGP, 2020).
- b. **Vehicle Pillar:** Encompassing vehicle suitability and the integration of advanced safety technologies aimed at mitigating vehicle-related risk factors.
- c. **Road Pillar:** Covering the physical condition and design of road infrastructure, including surface quality, signage, and environmental conditions.

While historically the Driver Pillar has garnered significant attention due to the predominance of human error in incidents, recent industry trends and research emphasize the critical role of the Vehicle Pillar in enhancing overall road safety outcomes. The adoption of vehicles equipped with advanced safety features—such as Anti-lock Braking Systems (ABS), Electronic Stability Control (ESC), and fatigue monitoring—can substantially reduce the frequency and severity of incidents by compensating for human limitations and errors (World Health Organization, 2018). By reinforcing the Vehicle Pillar, organizations can concurrently strengthen the Driver Pillar, creating a synergistic effect that minimizes risks more effectively.

To address these challenges, PDO implemented a comprehensive, integrated road safety management program known as "Darb Al Salama". This initiative harmonizes the three pillars of road safety—Safe Driver, Safe Vehicle, and Safe Road—underpinned by technology and digitalization to enable data-driven decision making and real-time risk mitigation. Central to this program is the renewal and enforcement of vehicle safety standards, which includes the mandatory integration of enhanced vehicle safety features across the PDO fleet and contractors.

The positive impact of advanced vehicle safety interventions on road safety has been well documented. For example, a study by Page and Hermitte (2001) demonstrated that the introduction of enhanced safety technologies, combined with enforcement measures such as speed cameras, contributed to a 48% reduction in road fatalities in France between 2000 and 2010. Similar findings have been reported globally, affirming that a multi-layered approach involving both engineering controls and behavioral interventions yields substantial safety benefits (Elvik, 2012; European Transport Safety Council, 2019).

This research paper focuses on the implementation of the renewed vehicle standards within PDO, highlighting how the adoption of advanced vehicle requirements has led to measurable improvements in road transport safety performance. It explores the challenges of integrating these standards across a diverse fleet and contractor base and presents empirical evidence demonstrating reductions in vehicle-related incidents following the program's rollout.

PDO New Vehicle Standard Specification SP2000V5

Vehicles equipped with advanced safety features have consistently demonstrated significant reductions in accident rates across multiple industries and geographies ([World Health Organization, 2018](#)). However, within the oil and gas sector, challenges remain in the rollout and implementation of these technologies, primarily due to the high cost of purchasing newer, safer vehicles and the logistical complexities involved in replacing existing contractor fleets. Many organizations adopt a phased approach prioritizing compliance in company-owned fleets, with gradual inclusion of contractor fleets over time.

In the case of Petroleum Development Oman (PDO), approximately 90% of the operational vehicle fleet is contractor-owned, which necessitates a strategic and collaborative approach in fleet safety adherence to with enhanced vehicle safety standards. However, management of this process requires time and dedicated resources. Therefore, planning for these resources and implementation were crucial for success of safe fleet achievements. Recognizing this, PDO introduced a comprehensive, renewed Road Safety Standard in early 2022 as part of its *Darb-Al Salama* initiative, with a particular focus on advancing vehicle safety requirements to achieve risk levels As Low As Reasonably Practicable (ALARP).

The new vehicle standard, designated SP2000V5, was rolled out with a phased implementation plan: all new contracts are required to comply with the standard within a 12-month grace period, providing contractors adequate time to procure or upgrade their fleets accordingly. For existing contracts, rejuvenation of the fleet and adherence to new standard features is expected based on the contract cycle as well as from some features that can be retrofitted over a three-year transition period. This approach balances operational realities with the imperative of improving road transport safety across PDO operations.

Vehicle Safety Critical Elements (VSCE)

To streamline the adoption of these requirements and ensure a clear, harmonized understanding among stakeholders, PDO introduced the concept of Vehicle Safety Critical Elements (VSCE). Drawing a parallel to process safety, where critical elements are components whose failure could lead to major incidents, PDO's VSCE are defined as the essential vehicle features whose absence or malfunction could directly contribute to a serious motor vehicle incident. These elements are mandatory and require continuous monitoring and maintenance to uphold vehicle safety integrity.

The following eight critical safety elements are mandated under the new standard:

1. Anti-Lock Braking System (ABS):

ABS prevents wheel lock-up during hard braking, allowing the driver to maintain steering control and avoid skidding. This technology significantly reduces stopping distances on slippery surfaces and is vital for maintaining vehicle stability during emergency maneuvers ([National Highway Traffic Safety Administration \[NHTSA\], 2018](#)).

2. Electronic Braking System (EBS):

An advancement over traditional braking systems, EBS electronically controls braking force distribution, improving braking efficiency and responsiveness, especially for heavy commercial vehicles. EBS enables coordinated braking across multiple axles and integrates with other safety systems to enhance overall vehicle control ([SAE International, 2017](#)).

3. **Electronic Stability Control (ESC):**

ESC actively monitors vehicle dynamics, detecting loss of traction or skidding, and automatically applies braking to individual wheels while adjusting engine torque to maintain vehicle stability. ESC is proven to reduce single-vehicle crashes and rollover incidents, making it essential for the challenging terrains encountered in PDO's operations ([NHTSA, 2014](#)).

4. **Rollover Prevention System (RPS):**

RPS specifically targets the prevention of rollovers by monitoring parameters such as lateral acceleration and vehicle tilt. It intervenes proactively to reduce rollover risk through brake actuation and engine power modulation, critical in off-road conditions and uneven surfaces common in oil field environments ([Euro NCAP, 2020](#)).

5. **Active Fatigue Detection Devices (AFDD):**

Fatigue is a major contributor to road crashes. DFMS use physiological and behavioral indicators — such as steering patterns, eye movement, and blink rates—to detect driver drowsiness in real-time and issue alerts to prevent accidents caused by fatigue ([Transport Research Laboratory, 2019](#)).

6. **Air Bags:**

Airbags provide a crucial passive safety measure by cushioning occupants in the event of a collision, reducing the risk of serious injuries and fatalities. Modern vehicles are equipped with multiple airbags, including front, side, and curtain airbags, to maximize occupant protection ([IIHS, 2020](#)).

7. **Tire Pressure Monitoring System (TPMS):**

TPMS continuously monitors tire pressure and alerts the driver to underinflated tires, which can impair vehicle handling and increase the risk of blowouts. Proper tire maintenance is essential for safe vehicle operation, especially under heavy load conditions prevalent in the oil and gas sector ([NHTSA, 2017](#)).

8. **In-Vehicle Monitoring System (IVMS):**

IVMS provide telematics data including vehicle speed, location, driver behavior, and operational parameters. These systems enable real-time monitoring, supporting proactive interventions to improve driver performance and fleet management. Advanced IVMS can integrate geofencing, speed regulation, and fatigue monitoring functions, thereby enhancing overall transport safety ([TNO, 2019](#)).

Implementing these Vehicle Safety Critical Elements across the PDO fleet represents a significant step forward in minimizing road transport risks. Ensuring that contractors comply with these requirements not only promotes safer driving behavior but also enhances vehicle reliability and resilience in the challenging operational environment of the Sultanate of Oman.

Safety Critical Elements Requirements for Different Vehicle Categories

The renewed vehicle safety standard SP2000V5 defines specific requirements tailored to different vehicle categories within PDO operations. These requirements were clearly communicated through the updated Road Safety Standard to ensure alignment across all operational fleets.

Table 2.1—Safety Critical Elements Requirements for Different Vehicle Categories

Sr. No.	Vehicle Category	ABS/EBS	TPMS	ESC/RPS	DFMS/ AFDD	Airbag	IVMS	Age
1	Sedan/Saloon	Yes	Yes	Yes	NA	Yes	Yes	8
2	SUV 4x4	Yes	Yes	Yes	NA	Yes	Yes	8
3	Pickup	Yes	Yes	Yes	NA	Yes	Yes	8
4	Ambulance	Yes	Yes	Yes	NA	Yes	Yes	10
5	3 Ton Canters	Yes	NA	NA	Yes	Yes	Yes	8
6	3 ton to 9 ton (Excluding Canter)	Yes	Yes	NA	Yes	NA	Yes	8
7	Minibus (<16 seats)	Yes	Yes	NA	Yes	NA	Yes	7
8	Light bus (<25 seats)	Yes	Yes	NA	Yes	NA	Yes	7
9	Heavy bus - Coach (>26seats)	Yes	Yes	Yes	Yes	NA	Yes	10
10	Heavy bus - Labor (>26seats)	Yes	Yes	NA	Yes	NA	Yes	10
11	HGV rigid (10T - 20T)	Yes	Yes	NA	Yes	NA	Yes	12
12	HGV rigid (>20T)	Yes	Yes	Yes	Yes	NA	Yes	12
13	HGV rigid with crane	Yes	Yes	NA	Yes	NA	Yes	12
14	HGV rigid tipper	Yes	Yes	NA	Yes	NA	Yes	12
15	HGV rigid tanker	Yes	Yes	Yes	Yes	NA	Yes	12
16	Prime mover	Yes	Yes	Yes	Yes	NA	Yes	12
17	Semitrailer-flatbed	Yes	NA	Yes	NA	NA	NA	15
18	Semitrailer-lowbed <=16T axel load	Yes	NA	Yes	NA	NA	NA	15
19	Semitrailer-lowbed >18T axel load	Yes	NA	NA	NA	NA	NA	15
20	Semitrailer-oilfield	Yes	NA	Yes	NA	NA	NA	15
21	Semitrailer-tipper	Yes	NA	Yes	NA	NA	NA	15
22	Semitrailer-tanker	Yes	NA	Yes	NA	NA	NA	15
23	Semitrailer-bulk tanker	Yes	NA	Yes	NA	NA	NA	15
24	Full Trailer	Yes	NA	Yes	NA	NA	NA	15
25	Bogie Axle Trailers	NA	NA	NA	NA	NA	NA	15
26	Specialized HGV rigid tucks	NA	Yes	NA	YES	NA	Yes	12
27	Plant machinery	NA	NA	NA	NA	NA	NA	NA
28	Mobile equipment	NA	NA	NA	NA	NA	NA	NA

NB:

- **Sedan/Saloon Cars** are permitted only within coastal areas due to operational risk profiles.
- **Active Fatigue Detection Devices (DFMS/AFDD)** are mandatory for vehicles operating at night or undertaking journeys exceeding 200 km one way.
- **Tire Pressure Monitoring Systems (TPMS)** are accepted when built in by the Original Equipment Manufacturer (OEM).

Certain specialized Heavy Goods Vehicle (HGV) rigid trucks (listed in [Appendix, Table 2.2](#)) are exempt from requirements relating to vehicle age, mileage, and core braking and stability systems such as ABS, EBS, and ESC. However, these vehicles must still comply with other critical safety elements including, IVMS, AFDD, underrun protection, and other specifications as per the OPAL regulatory standards.

Rollout Implementation Strategy

Introducing the SP2000V5 vehicle safety standard represented a significant shift with substantial financial and operational implications. Recognizing the potential resistance to change, PDO adopted a carefully designed and strategic rollout plan to ensure successful implementation across all stakeholders.

Alignment with Internal Stakeholders.

- **Senior Management Team:** Given the financial investment required to upgrade the vehicle fleet, securing senior leadership support was a priority. The business case emphasized long-term value, safety benefits, and operational excellence. Leadership endorsed the mandate, empowering Contracting and Procurement teams to negotiate and enhance contract rates to account for the cost of safer vehicles.
- **Integrated Supply Chain Team:** The new safety requirements were embedded into tender documentation and procurement processes. Commercial contracts were updated with the new safety clauses with clear timelines, ensuring contractors remained accountable for meeting standards.
- **Contract Holders and Owners:** Detailed roles and responsibilities were defined through a RASCI matrix aligned with the new vehicle specification. Training sessions equipped contract managers with the knowledge to manage the implementation effectively.
- **Health, Safety & Environment (HSE) Advisers:** Operationally, corporate and site embedded HSE advisers were mobilized to oversee the implementation, conduct periodic reviews, and track progress of compliance through KPIs, reinforcing a culture of safety vigilance.

Alignment with External Stakeholders.

- **Original Equipment Manufacturers (OEMs):** While many advanced safety features exist in the global automotive market, availability in the GCC region, particularly for buses and heavy trucks, was initially limited. PDO engaged extensively with OEMs to communicate the new specification and stimulate supply chain responses. This engagement accelerated OEM readiness, especially for heavy-duty trucks requiring Electronic Stability Control and electronic communication modules linking prime movers and trailers.
- **Oman Energy Association (OPAL):** The PDO vehicle standard aligns with the International Association of Oil & Gas Producers (IOGP) Vehicle Standard 365, but with more stringent implementation timelines. Collaboration with OPAL ensured harmonization of safety standards across Oman's oil and gas sector, culminating in the adoption of similar specifications within OPAL's updated road safety standard.
- **Other Oil & Gas Operators:** Coordination with major operators such as Shell, BP, Total, Occidental, and OQ facilitated industry-wide adoption. Shared contractors and common safety goals supported alignment across the sector.

Phased Rollout Implementation Strategy

Concerns raised during stakeholder engagement included potential business disruptions, financial exposure, and vehicle availability constraints. To mitigate these, PDO implemented a **phased rollout approach** balancing urgency with practicality:

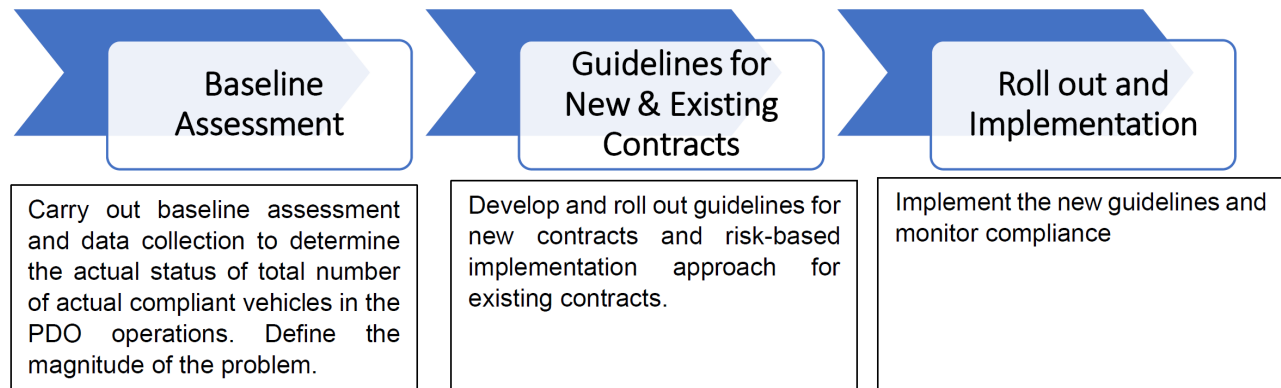


Figure 2.1—Roll Out Implementation Strategy

- New Contracts:** All new contracts awarded post-standard introduction mandate immediate alignment with SP2000V5 vehicle specifications. A transitional period given to enable the contractors to bring in the new fleet or retrofit (where technically was viable) their existing fleet. Usually, this period was from six months to one year transition period for fleet safety improvement, driven by the market availability and size of the fleet. Key challenges faced for the new contracts is related to fleet availability and supply chain restraints as the Gulf Standard Organization GSO specifications of the vehicles for GCC regions do not mandates some of the key safety features e.g ESC for heavy vehicles that has been a long time standard for all new vehicles in the European or US market, as early as 2017.
- Existing Contracts:** Current contracts were permitted to continue with non-compliant vehicles under strict mitigations controls and a defined transition period of three years. Vendors facing market availability challenges were required to submit mitigation plans and timelines.

This phased approach, combined with continuous collaboration with regulators and OEMs, ensured smooth adoption without operational interruptions.

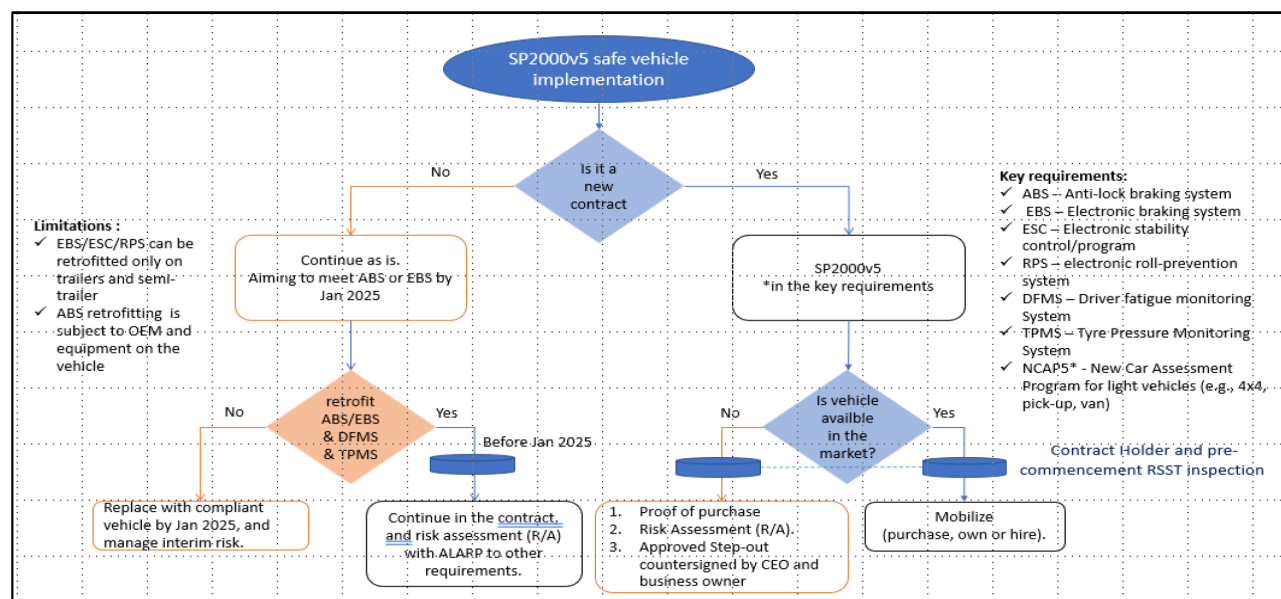


Figure 2.2—Flow chart for Safe Vehicle Implementation

Results and Outcomes

Safe Vehicle Implementation by Contracts

Since the rollout of the SP2000V5 standard in early 2022, PDO has achieved 38% of all vehicles with the latest safety technology, representing approximately 3,900 vehicles equipped with all required safety critical elements. Notably, 100% of newly awarded contracts are fully aligned with the new specifications, reflecting strong integration of safety requirements into procurement and contract management processes.

Vendors have shown increasing willingness to invest in compliant fleets, driven by revised commercial models and incentives built into contract structures. Engagement with energy industry regulatory authority (Oman Energy Association - OPAL) has also been instrumental, resulting in a harmonized vehicle safety standard across the Sultanate of Oman, aligned with international benchmarks such as the IOGP 365 Guidelines.

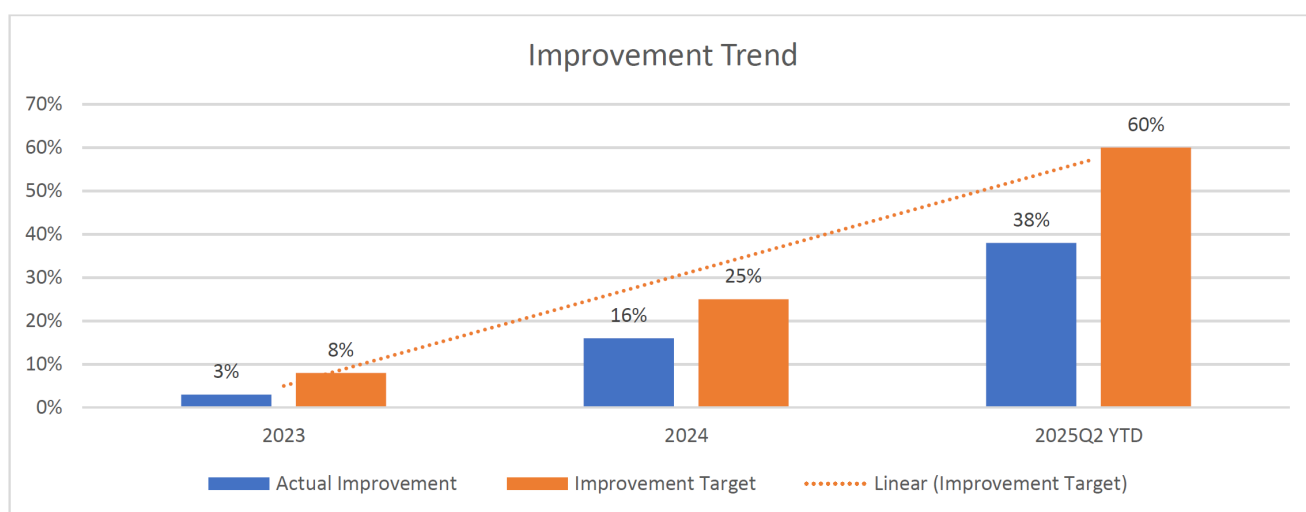


Figure 3.1—Vehicle safety improvement trend 2023 to 2025

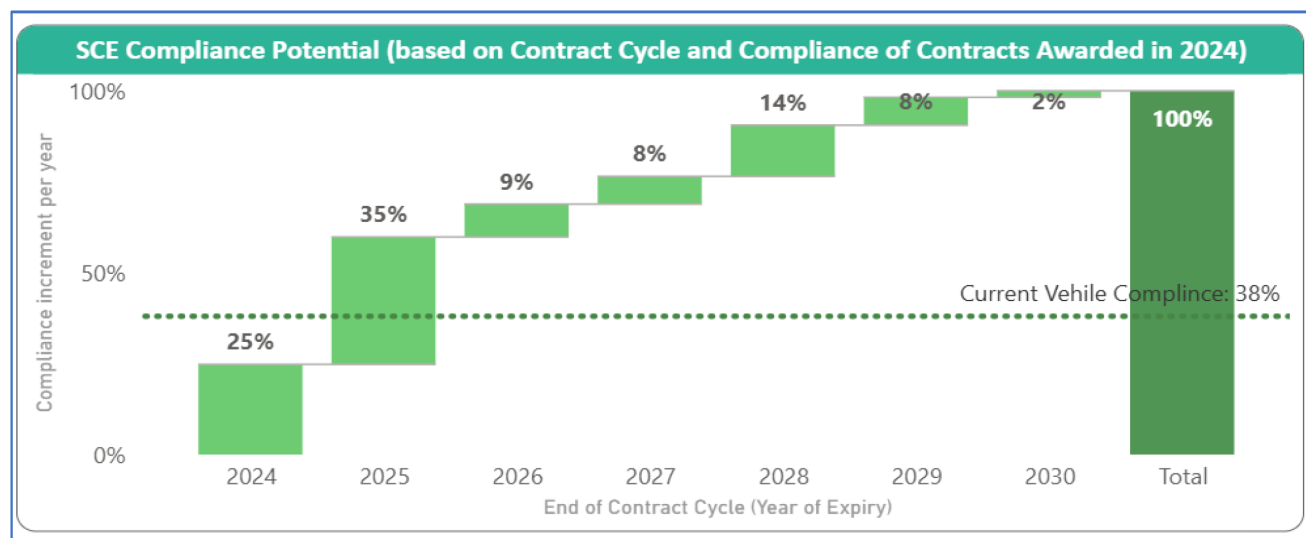


Figure 3.2—Vehicle Safety rejuvenation and improvement

Tracking Vehicle Safety through Critical Element Category

To support continuous monitoring and manage non-compliant vehicles safely, PDO developed specific Key Performance Indicators (KPIs) to track vehicle safety rejuvenation across each Safety Critical Element (SCE). Vehicles that are with older specifications due to contract maturity or market limitations are risk-assessed and managed under ALARP (As Low As Reasonably Practicable) protocols.

These vehicles are risk assessed and temporarily approved for operation under additional mitigation controls (e.g., speed caps, driving time limits, or fatigue monitoring devices). The vehicle safety rejuvenation and distribution across each SCE is illustrated in the dashboard below that has been developed in-house for visibility of the progress, as shows below:

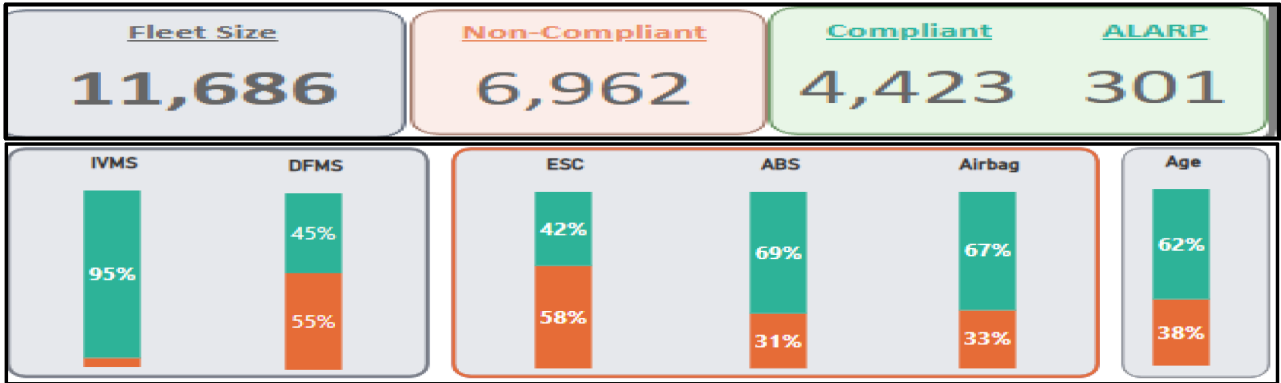


Figure 3.3—Example of PDO Vehicle Safety dashboard shows tracking of transition to modern safety vehicle feature

Since the introduction of SP2000V5, PDO has observed a year-on-year increase in both overall safe vehicle rejuvenation and adherence to individual safety elements. Ongoing efforts focus on converting ALARP-approved vehicles into upgraded and replaced vehicles by targeting all vehicles in PDO rejuvenated and the modern safety features by 2030.

Impact on PDO Road Safety Performance

The integration of SP2000V5 with other components of PDO's Darb-Assalama Road Safety Initiative has led to measurable improvements in road safety performance. Key highlights include:

- 70% improvement in Life-Saving Rules compliance, driven by data insights AFDD
- Significant reduction in work-related fatal and high-potential Motor Vehicle Incidents (MVIs) since 2021.
- Enhanced compliance monitoring through AFDD, including:
 - Real-time detection of fatigue, distraction, and mobile phone use.
 - Verification of seatbelt usage and Safe Journey Management (SJM) protocols.
- Deployment of vehicle safety technologies such as ESC, RPS, and NCAP has provided effective post-incident recovery, reducing the severity of accidents. Severe Motor Vehicle Crashes has reduced, and Incident investigations revealed new safety features either prevent incidents or had helped the occupants to withstand with no or minor injuries.

These improvements highlight the importance of combining engineering controls with behavioral monitoring tools to proactively manage road transport risk in high-exposure environments like PDO's desert operations.

Conclusion

The successful implementation of PDO's SP2000V5 vehicle safety standard illustrates how visionary leadership, multi-stakeholder collaboration, and a strategic phased rollout can overcome financial, logistical, and market-based challenges.

Despite initial resistance and OEM supply limitations—especially for heavy-duty vehicles—PDO has driven significant adoption of safety-enhancing technologies, setting a national benchmark for road transport safety. Through close coordination with OEMs, regulators (e.g., OPAL), and peer operators, PDO has aligned Oman's oil and gas sector with international safety best practices, including IOGP Guidelines.

More importantly, the program has resulted in tangible safety outcomes—notably the reduction in serious incidents and the transformation of contractor behavior through accountability, education, and monitoring.

As PDO targets 100% fleet rejuvenation to modern standards by 2030, this initiative serves as a model for other industries operating in high-risk, high-mobility environments, demonstrating how integrated safety strategies can deliver both operational and life-saving value.

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Appendix

Table 2.2—Specialized HGV rigid trucks excluded from the standard

Sr. No.	Specialized HGV rigid trucks (Tailor made)*
1	Acid tanker (considering the wall thickness)
2	Banana bulker
3	Batch mixer units' trailers
4	Blender trailers
5	Cement pumping trailers
6	Cement storage trailer
7	Cementing unit trailer
8	Chemical mixing tanks
9	Coil tubing unit
10	Coil tubing unit trailers
11	Data van trailer
12	Diesel trailer
13	Fluid pumping unit trailer (all shapes and designs)
14	FRAC pump trailer
15	FRAC sand bulker
16	FRAC tanks
17	Frac sander SDS trailer
18	Generator trailer
19	Hydration unit trailer
20	Hydraulic trucks
21	Iron trailer
22	Lab trailer
23	LAS trailer
24	Manifold trailer
25	Mobile service trucks (HDV)
26	Mount mover trailer
27	NCP unit
28	Nitrogen pumping unit trailer
29	Nitrogen unit transport trailers
30	Office trailer
31	Oilfield flatbed trailers
32	Oilfield winch trailer
33	Prime mover attached with coil tubing unit
34	Prime mover with powerpack
35	Raw cement tankers
36	Rig move camp mounted trailer
37	Sand catcher SMS trailer

Sr. No.	Specialized HGV rigid tucks (Tailor made)*
38	SLICKLINE truck unit
39	Slickline unit truck
40	SPM/trailer
41	Tea room trailer
42	Tech command center trailer
43	Twin pump
44	Well test unit trailer
45	Wireline truck unit